Category	Algorithm / Structure	Average Time	Worst Time	Space Complexity
Search	Binary Search	O(logN)	O(logN)	O(1)
Search	Quickselect (Hoare's)	O(N)	O(N2)	O(1) iterative, O(logN) recursive
Search	Depth-First Search (DFS)	O(V+E)	O(V+E)	O(V) for call stack
Search	Breadth-First Search (BFS)	O(V+E)	O(V+E)	O(V) for queue
Array	Two Pointers	O(N)	O(N)	O(1)
Array	Sliding Window	O(N)	O(N)	O(1) or O(K) for hash map
Array	Prefix Sum (Build / Query)	O(N)/O(1)	O(N)/O(1)	O(N)
Sorting	Quicksort	O(NlogN)	O(N2)	O(logN) for call stack
Sorting	Merge Sort	O(NlogN)	O(NlogN)	O(N)
Sorting	Heapsort	O(NlogN)	O(NlogN)	O(1)
Graph	Topological Sort (Kahn's)	O(V+E)	O(V+E)	O(V)
Graph	Dijkstra (with Min-Heap)	O((V+E)logV)	O((V+E)logV)	O(V)
Graph	Bellman-Ford	O(V×E)	O(V×E)	O(V)
Graph	Floyd-Warshall	O(V3)	O(V3)	O(V2)
Sets	Union-Find (Path Comp + Rank)	O(α(N))	O(logN)	O(N)
Trees	Trie (Insert / Search)	O(L)	O(L)	O(N×L)
Heaps	Priority Queue (Insert / Pop)	O(logN)	O(logN)	O(N)
Heaps	Heapify (Build Heap from array)	O(N)	O(N)	O(1) in-place
DP	1D Dynamic Programming	O(N)	O(N)	O(N) array or O(1) variables
DP	2D Dynamic Programming	O(N×M)	O(N×M)	O(N×M) matrix or O(M) row
Search	Backtracking (Subsets/Combos)	O(2N)	O(2N)	O(N) for call stack
Search	Backtracking (Permutations)	O(N!)	O(N!)	O(N) for call stack
Search	Cycle Detection (Floyd's Tortoise & Hare)	O(N)	O(N)	O(1)
Array	Monotonic Stack	O(N)	O(N)	O(N)
Array	Boyer-Moore Majority Vote	O(N)	O(N)	O(1)
Graph / MST	Kruskal's Algorithm	O(ElogE)	O(ElogE)	O(V)
Graph / MST	Prim's Algorithm (with Min-Heap)	O((V+E)logV)	O((V+E)logV)	O(V)
Graph	A* Search (with Heuristic)	Depends on heuristic	O(bd)	O(bd)
Graph	Bipartite Checking (Graph Coloring)	O(V+E)	O(V+E)	O(V)
Trees	Segment Tree (Build)	O(N)	O(N)	O(N)
Trees	Segment Tree (Query / Update)	O(logN)	O(logN)	O(1) after build
Trees	Fenwick Tree / BIT (Build)	O(NlogN)	O(NlogN)	O(N)
Trees	Fenwick Tree / BIT (Query / Update)	O(logN)	O(logN)	O(1) after build
Trees	Lowest Common Ancestor (Binary Lifting)	O(logN) query	O(logN) query	O(NlogN) preprocessing
Strings	KMP (Knuth-Morris-Pratt) Pattern Matching	O(N+M)	O(N+M)	O(M) for LPS array
Strings	Rabin-Karp (Rolling Hash)	O(N+M)	O(N×M)	O(1)
Math	Sieve of Eratosthenes (Primes)	O(Nlog(logN))	O(Nlog(logN))	O(N)
Math	Euclidean Algorithm (GCD)	O(log(min(A,B)))	O(log(min(A,B)))	O(1)